

# Digital Forensics Laboratory

EXPERIMENT NO: 02

DATE: August 9, 2026

TITLE: Recover deleted or damaged files from a storage device using TestDisk

## 1. AIM

To recover deleted files and repair a missing or corrupted partition from a storage device using the TestDisk data recovery command-line utility.

## 2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** TestDisk (v7.3-WIP Data Recovery Utility)
- **Hardware:** Target Storage Device

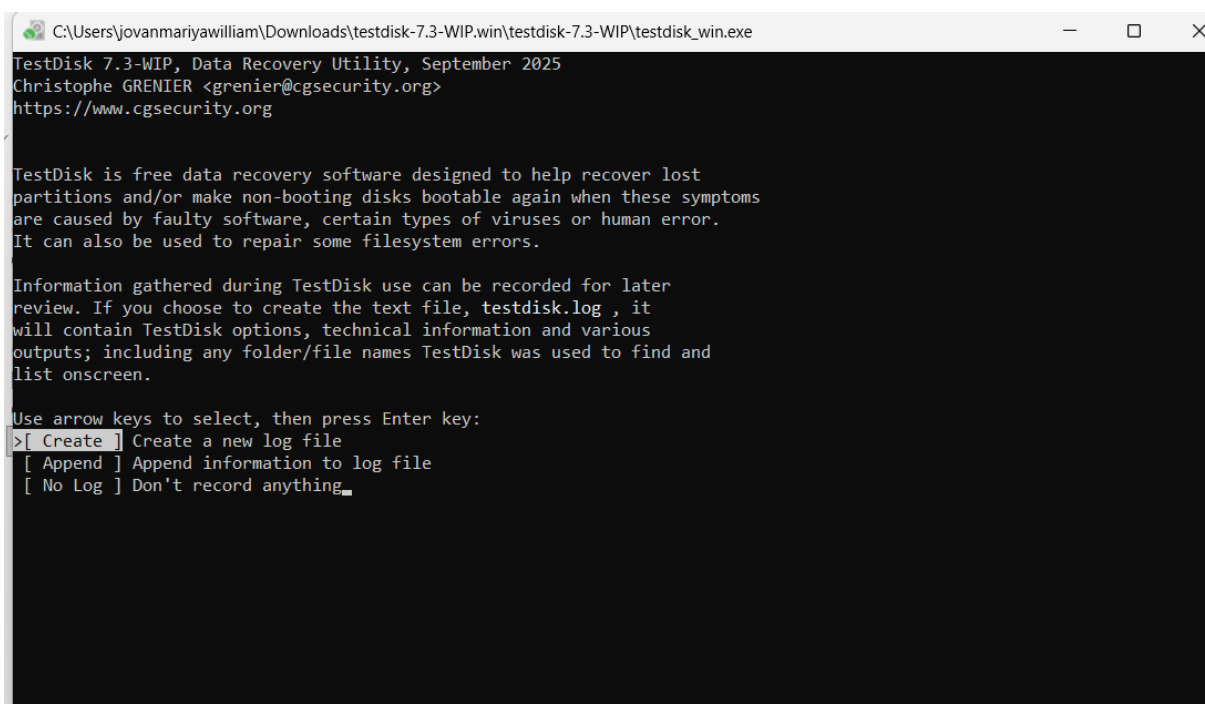
## 3. THEORY

TestDisk is a powerful, open-source data recovery software primarily designed to help recover lost partitions and make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses, or human error (such as accidentally deleting a Partition Table). It can also be utilized to undelete files from FAT, exFAT, NTFS, and ext2 file systems. TestDisk queries the BIOS or the OS in order to find the hard drives and their characteristics, performs a quick check of the disk structure, and looks for lost partitions or deleted files.

## 4. PROCEDURE

### Step 1: Log Creation & Disk Selection

1. Open the testdisk\_win.exe application as an Administrator.
2. **Log Creation:** Choose [Create] to instruct TestDisk to create a log file containing technical information and messages. Press Enter to proceed.



```
C:\Users\jovanmariyawilliam\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log, it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything
```

Figure 1: Initial TestDisk dashboard prompting for log creation

**3. Disk Selection:** All connected hard drives will be detected and listed. Use the up/down arrow keys to select the target hard drive containing the lost partitions or deleted files (e.g., Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w). Press Enter to Proceed.

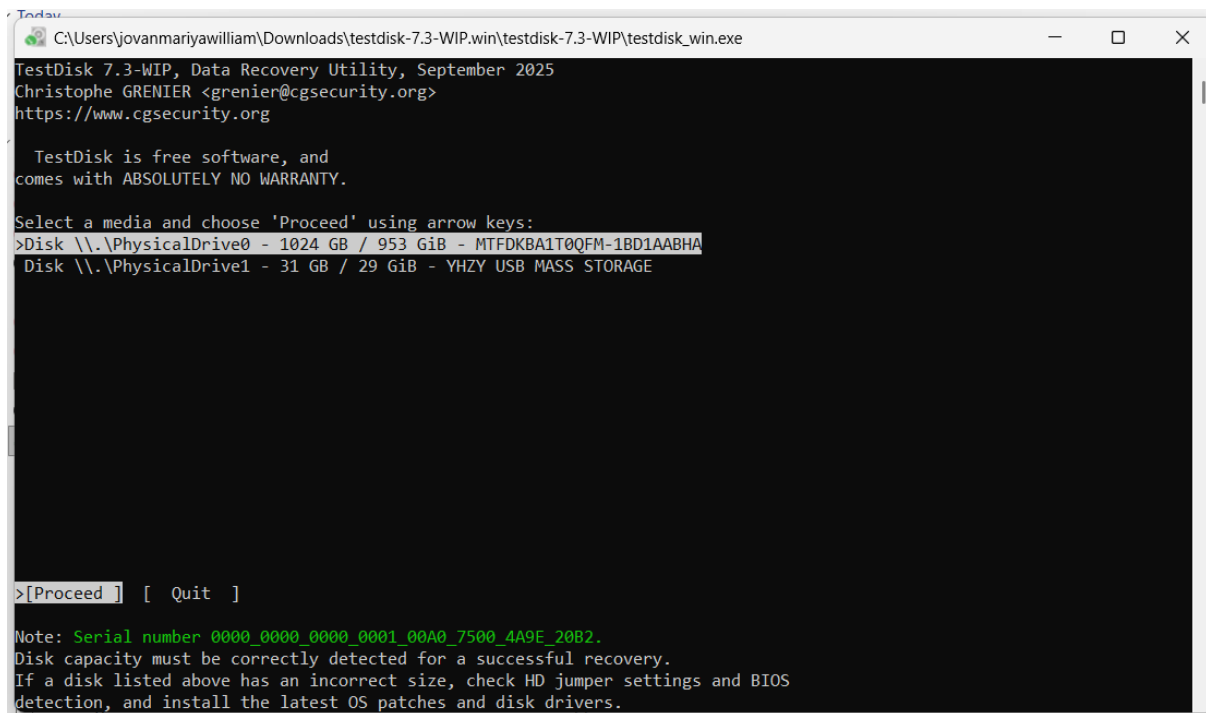


Figure 2: Selecting the physical storage drive in TestDisk

## Step 2: Partition Table Type Selection

1. TestDisk displays the available partition table types.
2. Select the partition table type. Usually, the default value is the correct one as TestDisk auto-detects the partition table type (e.g., [Intel] Intel/PC partition). Press Enter to Proceed.

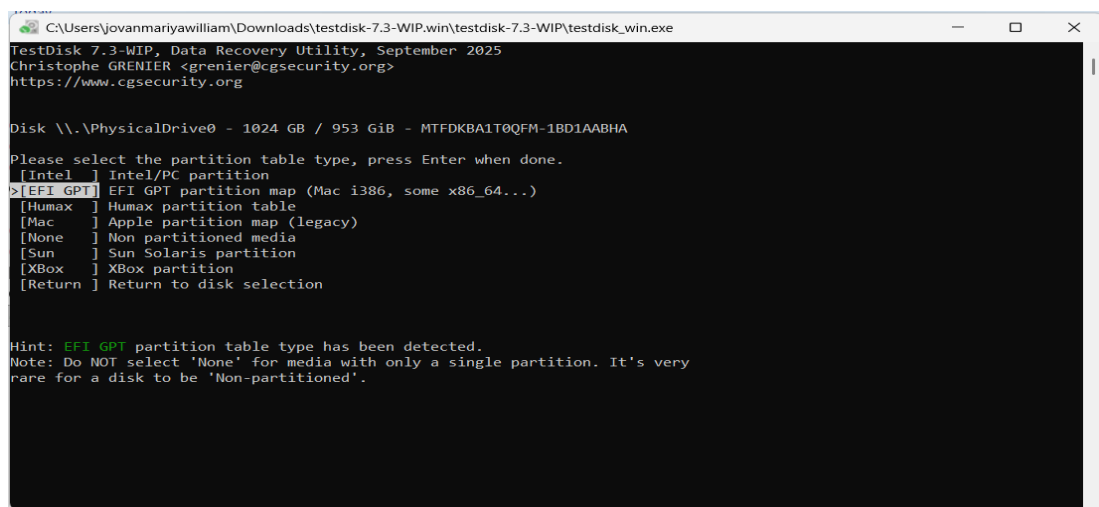
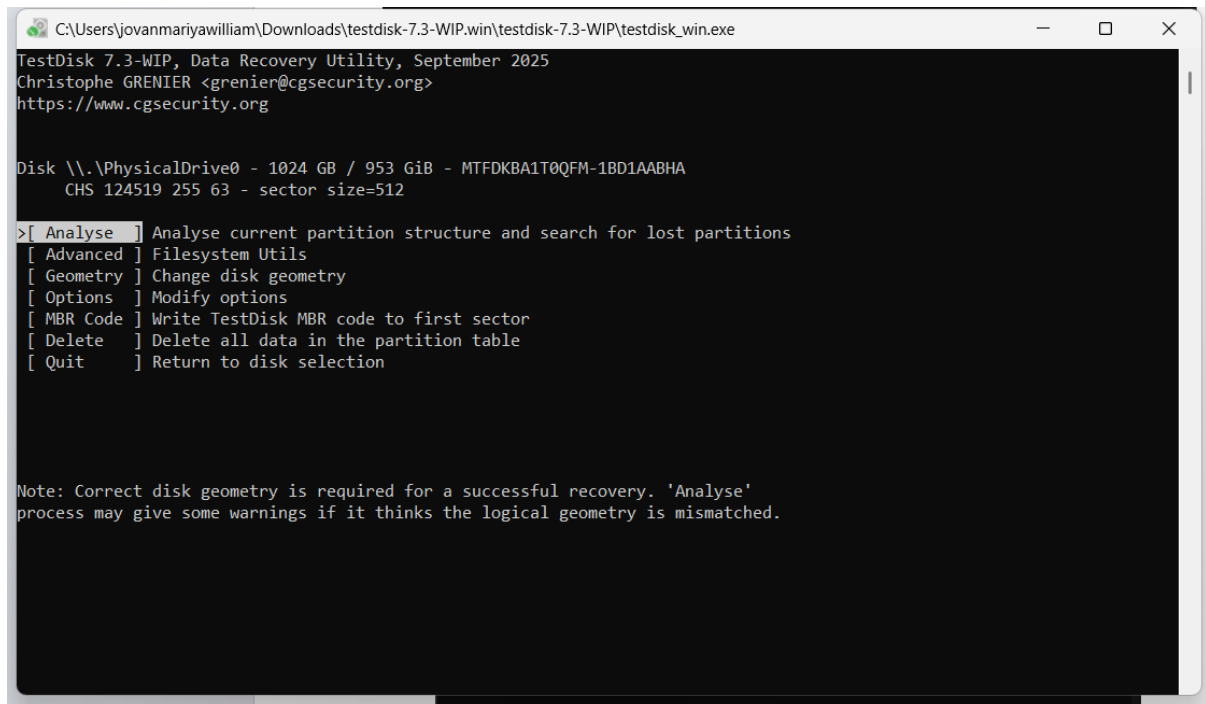


Figure 3: Selecting the Intel/PC partition table type

### Step 3: Current Partition Table Status

1. TestDisk displays the main menu options. Use the default menu [Analyse] to check your current partition structure and search for lost partitions. Confirm at Analyse with Enter.

A screenshot of the TestDisk 7.3-WIP application window. The title bar shows the file path: C:\Users\jovanmariyawilliam\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk\_win.exe. The main text area displays the following information:  
TestDisk 7.3-WIP, Data Recovery Utility, September 2025  
Christophe GRENIER <grenier@cgsecurity.org>  
<https://www.cgsecurity.org>  
  
Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - MTFDKBA1T0QFM-1BD1AABHA  
CHS 124519 255 63 - sector size=512  
  
The menu options are listed as follows:  
> [ Analyse ] Analyse current partition structure and search for lost partitions  
[ Advanced ] Filesystem Utils  
[ Geometry ] Change disk geometry  
[ Options ] Modify options  
[ MBR Code ] Write TestDisk MBR code to first sector  
[ Delete ] Delete all data in the partition table  
[ Quit ] Return to disk selection  
  
A note at the bottom states: Note: Correct disk geometry is required for a successful recovery. 'Analyse' process may give some warnings if it thinks the logical geometry is mismatched.

2. The current partition structure is listed. Examine the structure for missing partitions and errors (e.g., Invalid NTFS boot, missing logical partitions). Confirm at [Quick Search] to proceed.

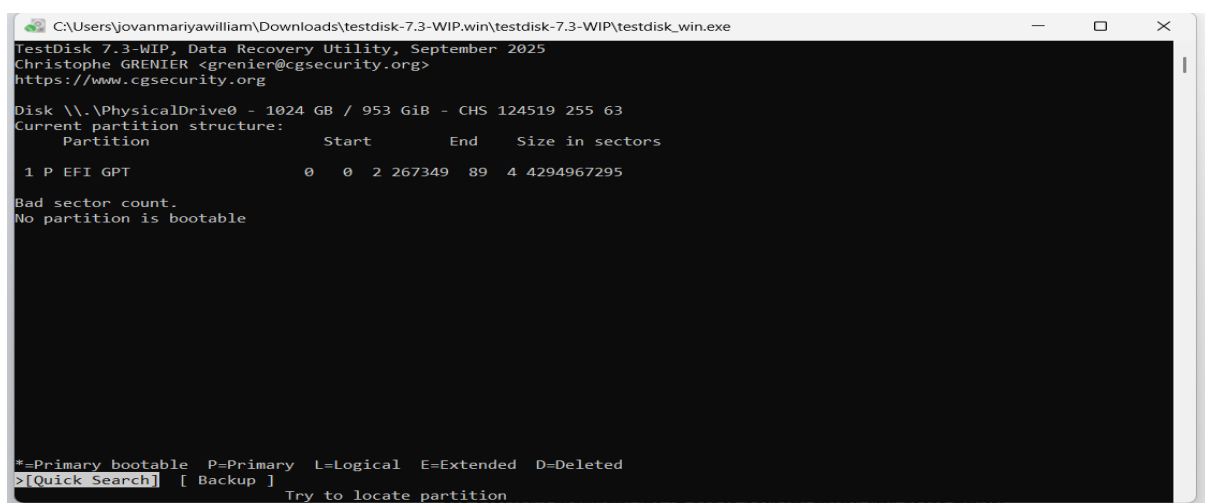
A screenshot of the TestDisk 7.3-WIP application window showing the current partition structure. The title bar is the same as the previous screenshot. The main text area displays:  
TestDisk 7.3-WIP, Data Recovery Utility, September 2025  
Christophe GRENIER <grenier@cgsecurity.org>  
<https://www.cgsecurity.org>  
  
Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - CHS 124519 255 63  
Current partition structure:  
Partition Start End Size in sectors  
1 P EFI GPT 0 0 2 267349 89 4 4294967295  
  
Below the table, it says: Bad sector count.  
No partition is bootable  
  
At the bottom, there is a legend: #=Primary bootable P=Primary L=Logical E=Extended D=Deleted  
The prompt >[Quick Search] [ Backup ] is shown, with the instruction Try to locate partition below it.

Figure 5: Initiating a Quick Search for lost partitions

#### Step 4: Deeper Search (If Required)

1. If the quick search does not find the required partition or if partitions overlap (status 'D' for deleted), highlight the [Deeper Search] menu and press Enter.
2. Deeper Search will scan each cylinder for FAT32 backup boot sectors, NTFS backup boot superblocks, etc., to detect more partitions.

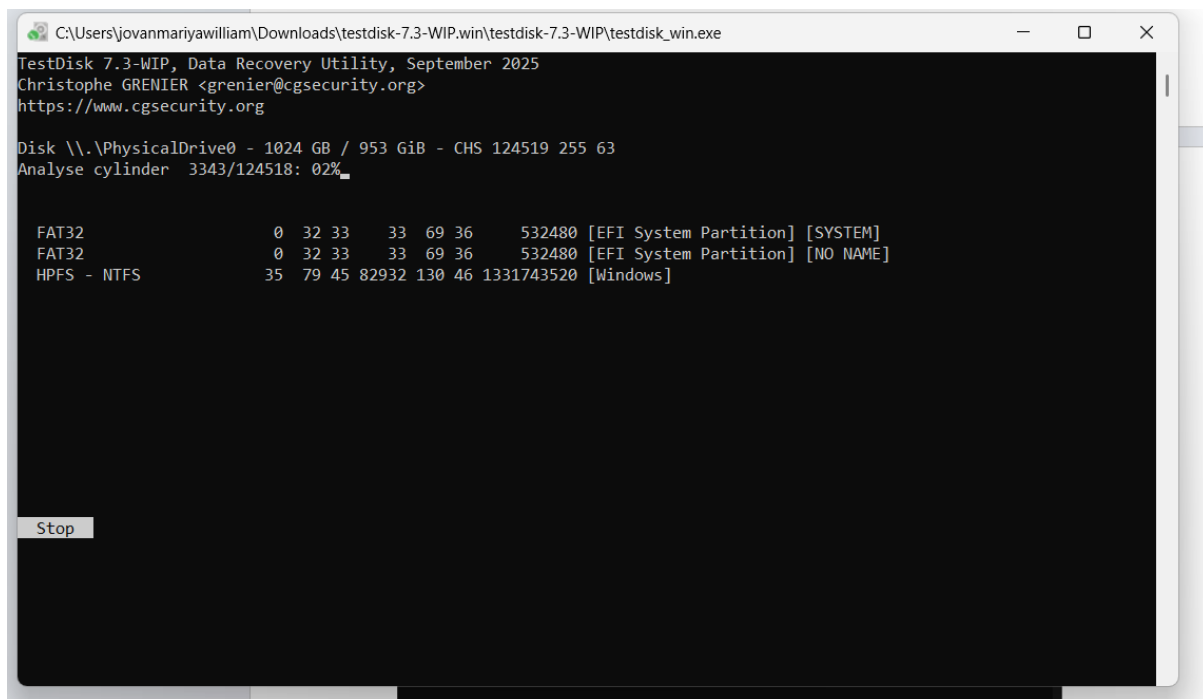


Figure 7: Deeper Search actively scanning disk cylinders

#### Step 5: Identifying and Accessing the Correct Partition

1. Once the search is complete, highlight the correct partition you wish to recover or extract files from (e.g., >L FAT32 LBA).
2. Press the p key to list the files inside that partition.

#### Step 6: Recovering / Copying Deleted Files

1. In the file listing, navigate through the folders using arrow keys. Files listed in red indicate deleted entries.
2. Highlight the specific deleted file you want to recover (e.g., images (1).jpg).

```

C:\Users\jovanmariyawilliam\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - CHS 124519 255 63
Partition      Start          End      Size in sectors
>* FAT32        0 32 33      33 69 36      532480 [EFI System Partition] [SYSTEM]
D HPFS - NTFS   35 79 45 82932 150 34 1331744768 [Windows]
D HPFS - NTFS   35 79 45 124445 219 62 1998655488 [Windows]
D Linux         15256 237 22 31966 35 5 268433408
D Linux         15259 89 63 31968 142 46 268433408
D Linux         17630 165 8 34339 217 54 268433408
D Linux         17708 234 63 34418 32 46 268433408
D Linux         17796 30 27 34505 83 10 268433408
D Linux         17798 40 35 34507 93 18 268433408
D Linux         17812 208 61 34522 6 44 268433408
D Linux         17846 121 4 34555 173 50 268433408
D Linux         17850 173 52 34559 226 35 268433408
D Linux         18183 7 1 34892 59 47 268433408
D Linux         26359 251 5 43069 48 51 268433408
D Linux         32005 135 3 48714 187 49 268433408
D Linux         69526 243 30 86236 41 13 268433408
D Linux         69538 141 44 86247 194 27 268433408
D Linux         69541 91 55 86250 144 38 268433408
Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 272 MB / 260 MiB

```

3. Press c to copy the current selected file (or C to copy all selected files).
  4. You will be prompted to select a destination directory to save the recovered file.
- Navigate to the desired local folder and press C when the destination is correct.
5. TestDisk will display a confirmation message indicating a successful recovery (e.g., "Copy done! 1 ok, 0 failed").

### Step 7: Partition Table and Boot Sector Recovery (Optional)

1. Press q to quit the file listing and return to the partition selection.
2. Use the left/right arrow keys to change the status of a partition from D (Deleted) to L (Logical) or P (Primary) to recover the partition itself. Press Enter.
3. If all partitions are correctly listed, confirm at [Write] with Enter, press y to confirm, and then OK.
4. If a boot sector is damaged, use the [Backup BS] option to copy the backup boot sector over the damaged one.
5. Exit TestDisk and reboot the computer to access the restored partition data.

## 6. RESULT

The TestDisk data recovery utility was successfully executed to analyze the physical drive's geometry and partition tables. Lost partitions were successfully identified through deeper searching, and deleted files were effectively navigated, selected, and copied (recovered) to a secure local destination.